



Modernize Infrastructure and Applications with Google Cloud

This study guide covers the materials from the fourth course of the six-course series for the Cloud Digital Leader certification.

Certification exam focus

The Cloud Digital Leader certification exam is designed to assess knowledge of cloud literacy, common business use cases, and how cloud solutions and the capabilities of Google Cloud products support an enterprise.

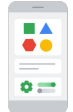
Cloud migration terms

- **Compute:** The ability of a machine to process information, like handling tasks such as storage, retrieval, and analysis.
- **Discovery and assessment:** The initial phase of a migration project where an organization inventories its existing on-premises environment.
- **Workload:** A specific app, service, or capability that can run in the cloud or on-premises.



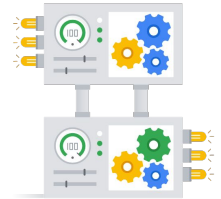
- **Retired workload:** When a workload is removed from a platform.
- **Retained workload:** When a workload is intentionally kept on a platform.
- **Rehosted workload (lift and shift):** When a workload is migrated to the cloud without changing anything in its code or architecture.
- **Replatformed workload (move and improve):** When a workload is migrated to the cloud after making changes to its code or architecture.
- **Reimagine:** The process of rethinking how an organization uses technology to achieve its business goals.

App: A computer program or software that helps users do something.



Cloud infrastructure terms

- **Virtual machines (VMs):** A software emulation of a physical computer that runs its own OS using shared hardware resources.
- **Spot VMs:** Discounted, excess compute resources that Google Cloud can reclaim at any time.
- **Managed services:** Cloud services where the provider handles maintenance and operations, letting users focus entirely on the app.
- **Containerization and containers:** A lightweight virtualization method that packages code and its dependencies to ensure consistent performance across different environments.



Compute Engine: Google Cloud's Infrastructure as a Service (IaaS) product that lets users create and run VMs on Google infrastructure.

- No upfront investments.
- Each VM functions as a complete OS.



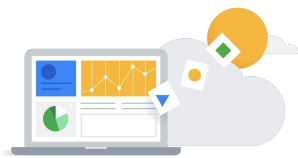


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Cloud architecture terms



Microservices: When an app is structured as a collection of small, autonomous services that communicate over a network, rather than one large, monolithic block of code.



Serverless computing: An architectural pattern where the developer focuses on the logic while the execution environment is abstracted away.



Load balancing: Distributing incoming network traffic across a group of backend servers to ensure no single server becomes overwhelmed.

Kubernetes: An open-source system for automating the deployment, scaling, and management of containerized apps.



Google Kubernetes Engine (GKE):

A Google-hosted managed Kubernetes service in the cloud. The GKE environment consists of multiple Compute Engine instances, grouped to form a cluster.

Google Cloud serverless computing products



Cloud Run

A fully managed environment for running containerized apps that can answer multiple events at the same time.



Cloud Run functions

A platform for hosting simple, single-purpose functions that are attached to events one at a time that are emitted from your cloud infrastructure and services.



App Engine

A service to build and deploy web apps.

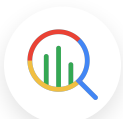
Benefits of modern cloud app development

- Microservices architecture
- Deployment with managed services
- Pay-as-you-go pricing
- Scalability
- Highly available and resilient
- Monitoring and management tools

Google Cloud products supported on multicloud and hybrid cloud environments



AlloyDB Omni



BigQuery Omni



GKE Enterprise



Cloud SQL



Looker





Modernize Infrastructure and Applications with Google Cloud

Application programming interface (API):

A set of protocols and definitions that enables distinct software apps to interface and exchange data. It serves as a standardized translation layer, ensuring predictable, secure communication between systems.



Apigee API Management

Google Cloud's API management service to operate APIs with enhanced scale, security, and automation.

- It helps organizations secure their APIs.
- It tracks and analyzes API usage.
- It helps with developing and deploying APIs.
- It offers API versioning, API documentation, and even API throttling.